

CS & IT ENGINEERING



Digital Logic

Combinational Circuits

Lecture No. 01



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Recap of Previous Lecture



Topic



K-maps

I

PI

EPI

$\left\{ \begin{array}{l} SPI \\ RPI \end{array} \right\} \rightarrow \text{Selected PI}$

$\rightarrow \text{Reduced PI.}$

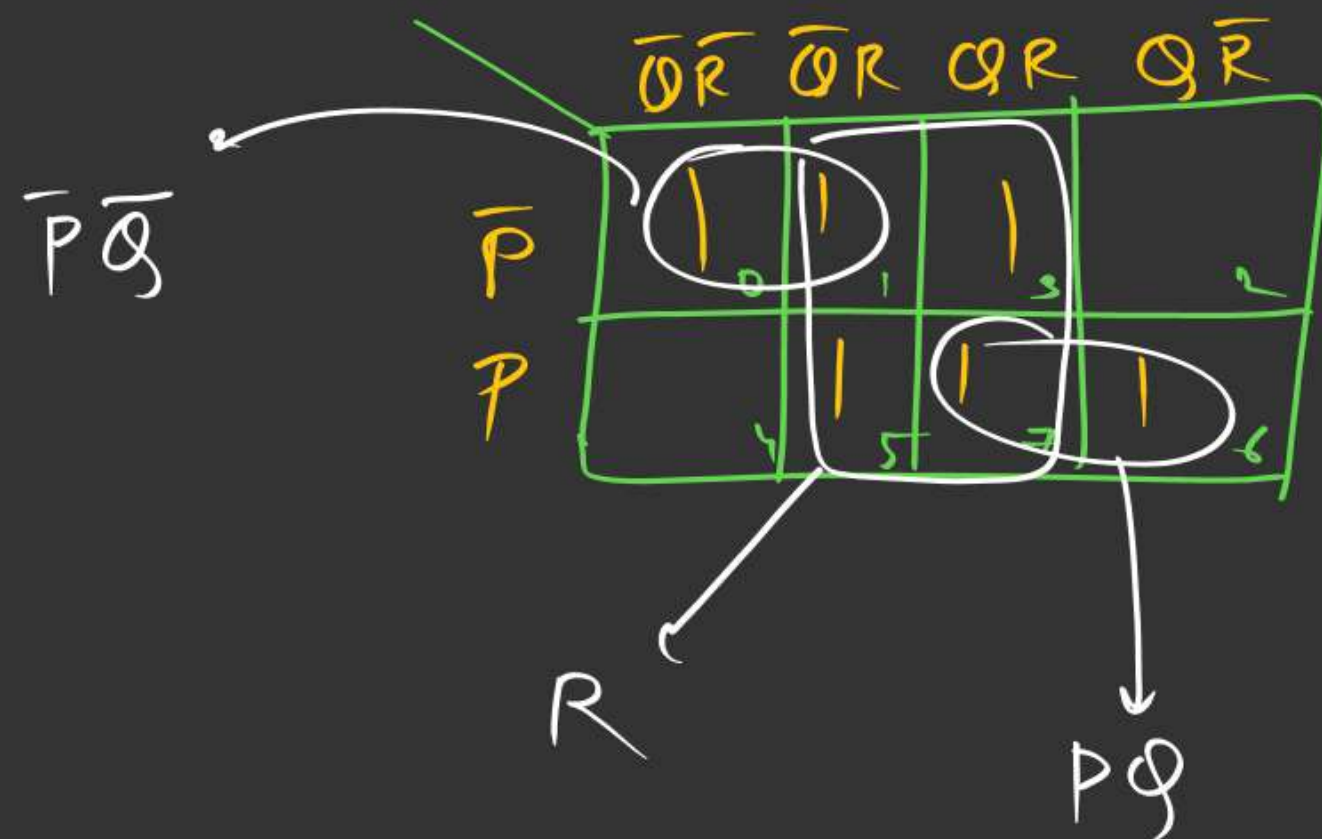
Topics to be Covered



Topic



- V. Imp Questions + Concept
- Duality
- Intro to Combinational ckt.

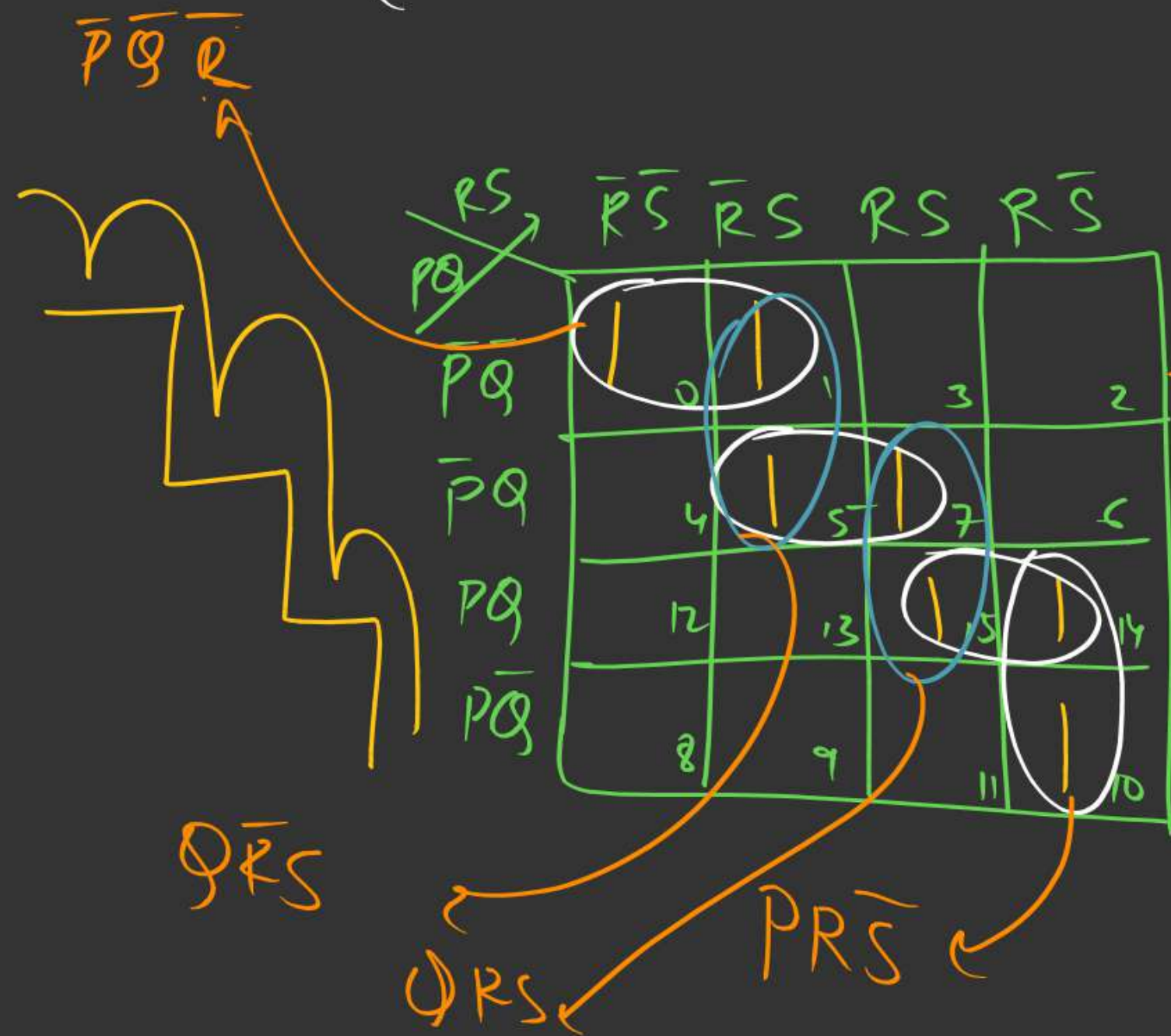


$$\begin{aligned}
 I &= 6 \\
 PI &= 3 \\
 EPI &= 3 \\
 SPI &= 0 \\
 RPI &= 0
 \end{aligned}$$

$$PI = \{ \bar{P}\bar{Q}, R, PQ \}$$

$$f = (\bar{P}\bar{Q} + PQ + R)$$

(Q2) $f(P, Q, R, S) = \sum m(0, 1, 5, 7, 10, 14, 15)$



$f = \bar{P}\bar{Q}\bar{R} + \bar{P}Q\bar{R} + \bar{P}\bar{Q}S + \bar{P}QR + \bar{P}QS + PR\bar{S}$

EPI

$I = 7$

$PI = 6$

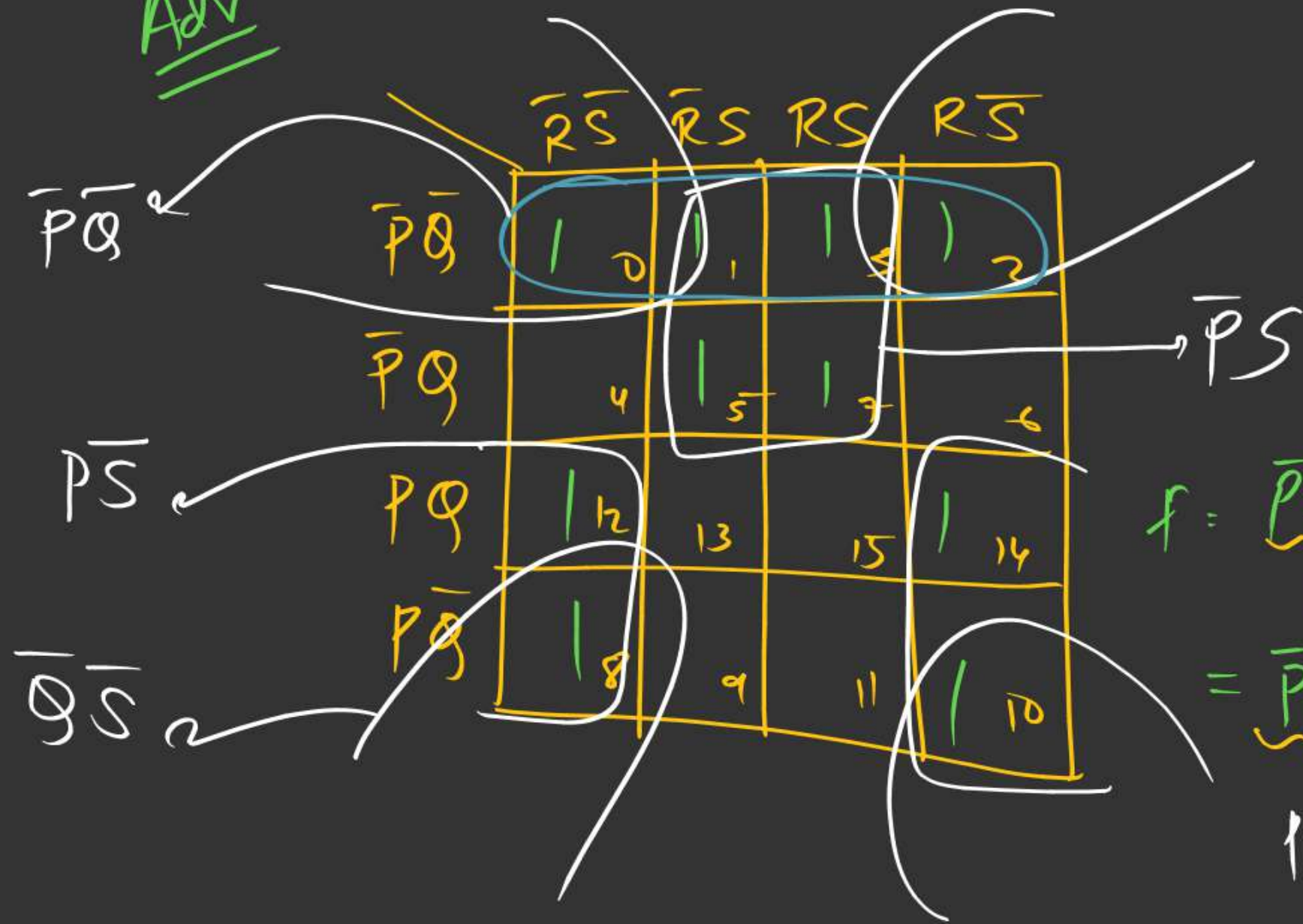
$EPI = 2$

$SPI = 2$

$KPI = 2$

(9) $f(P, Q, R, S) = \sum m(0, 1, 2, 3, 5, 7, 8, 10, 12, 14)$

Adv



$I = 10$

$PI = 4$

$EPI = 2 \{ \bar{P}S, PS \}$

$SPI = 1$

$RPI = 1$

$f = \bar{P}S + P\bar{S} + \bar{Q}\bar{S}$
OR

$= \bar{P}S + P\bar{S} + \bar{P}\bar{Q}$

$PI = \{ \bar{P}\bar{Q}, P\bar{S}, \bar{Q}\bar{S}, \bar{P}S \}$

$$(Q) \quad F(P, Q, R, S) = \sum m(0, 5, 6, 9, 10, 12)$$

	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$
$\bar{P}\bar{Q}$	1 0			
$\bar{P}Q$		1 4		1 6
$P\bar{Q}$	1 12			
PQ		1 8		1 10



$$I = 6$$

$$PI = 6$$

$$EPI = 6$$

$$SPI = 0$$

$$RPI = 0$$

Duality

(voltage)

$1 \leftrightarrow 0$
 $0 \leftrightarrow 1$
i/p & o/p
(Relative)

Logic

+ve Logic

-ve Logic

$1 \rightarrow 15V$
 $0 \rightarrow 5V$ } +ve

$1 \rightarrow 7V$
 $0 \rightarrow 0V$ } +ve

$1 \rightarrow 0V$
 $0 \rightarrow -7V$ } +ve

$1 \rightarrow -7V$
 $0 \rightarrow -10V$ } +ve

$-7 > -10$
1 0

1 $\rightarrow$ High
0 $\rightarrow$ Low

0 $\rightarrow$ High
1 $\rightarrow$ Low

Dual

$1 \rightarrow -10V$
 $0 \rightarrow -15V$ } +ve

$10 > 7$
0 1

$1 \rightarrow 7V$
 $0 \rightarrow 10V$ } -ve

$1 \rightarrow 0V$
 $0 \rightarrow 7V$ } -ve

$1 \rightarrow -7V$
 $0 \rightarrow -2V$ } -ve

Dual

OR

eg:-

(+ve logic)

AND

$X = P \cdot Q$

P	Q	X
0	0	0
0	1	0
1	0	0
1	1	1

↔

P	Q	X
1	1	1
1	0	1
0	1	1
0	0	0

$X = P + Q$

Is $\overline{P+Q} = P \cdot Q$?

No

But AND & OR are dual of each other.

Properties/Concepts:

pos logic $\xrightarrow{\text{dual}}$ neg logic

Duals $\rightarrow$

① $1 \xrightarrow{\text{dual}} 0$

② $\text{AND} \longleftrightarrow \text{OR}$

③ $\cdot \longleftrightarrow +$

④ $\text{NAND} \longleftrightarrow \text{NOR}$

★ ⑤ $\text{XOR} \longleftrightarrow \text{XNOR}$

⑥ $\text{Buffer} \longleftrightarrow \text{Buffer}$

⑦ $\text{NOT} \longleftrightarrow \text{NOT}$

$$\left[\begin{array}{l} \text{AND} \longleftrightarrow \text{OR} \\ \overline{\text{AND}} \longleftrightarrow \overline{\text{OR}} \\ \text{NAND} \longleftrightarrow \text{NOR} \end{array} \right]$$

} variables/ips remain same

$$0 \longleftrightarrow 1$$

$$\left\{ \begin{array}{l} A \longleftrightarrow A \\ \bar{A} \longleftrightarrow \bar{A} \end{array} \right\}$$

Self-Dual

$$f^D = f$$

$(f^D \rightarrow \text{dual of } f)$

$$f = p$$

$$f^D = p$$

$$p \cdot 1$$

$$\downarrow$$

$$p + 0 = p$$

(Q)

$$\overline{F} = \overline{P \cdot Q}$$

$$= \overline{P} + \overline{Q} \neq F^D$$

$$\begin{array}{l} F = P \cdot Q \\ \downarrow \quad \quad \downarrow \\ \underline{\underline{F^D = (P + Q)}} \end{array}$$

$$(Q) \quad f = P \cdot Q + RS + \overline{A}B$$

$$f^D = (P+Q) \cdot (R+S) \cdot (\overline{A}+B)$$

(Q)

$$f = \overline{PQR + ABC}$$

$$F^D = \overline{(P+Q+R) \cdot (A+B+C)}$$

$$(Q) \quad F = \overline{P+Q} \longrightarrow \underline{NOR}$$

$$F^D = \overline{P \cdot Q} \longrightarrow \underline{NAND}$$

NOR $\longleftrightarrow$ NAND

$$(8) \quad F = \bar{P}Q + P\bar{Q} \longrightarrow \text{XOR}$$

$$F^D = (\bar{P} + Q) \cdot (P + \bar{Q})$$

$$= \cancel{\bar{P} \cdot P} + \bar{P} \cdot \bar{Q} + Q \cdot P + \cancel{Q \cdot \bar{Q}}$$

$$= \bar{P}\bar{Q} + PQ \longrightarrow \text{XNOR}$$

$$[\text{XOR} \xleftrightarrow{\text{dual}} \text{XNOR}]$$

$$\boxed{(f^D)^D = f}$$

(Q)

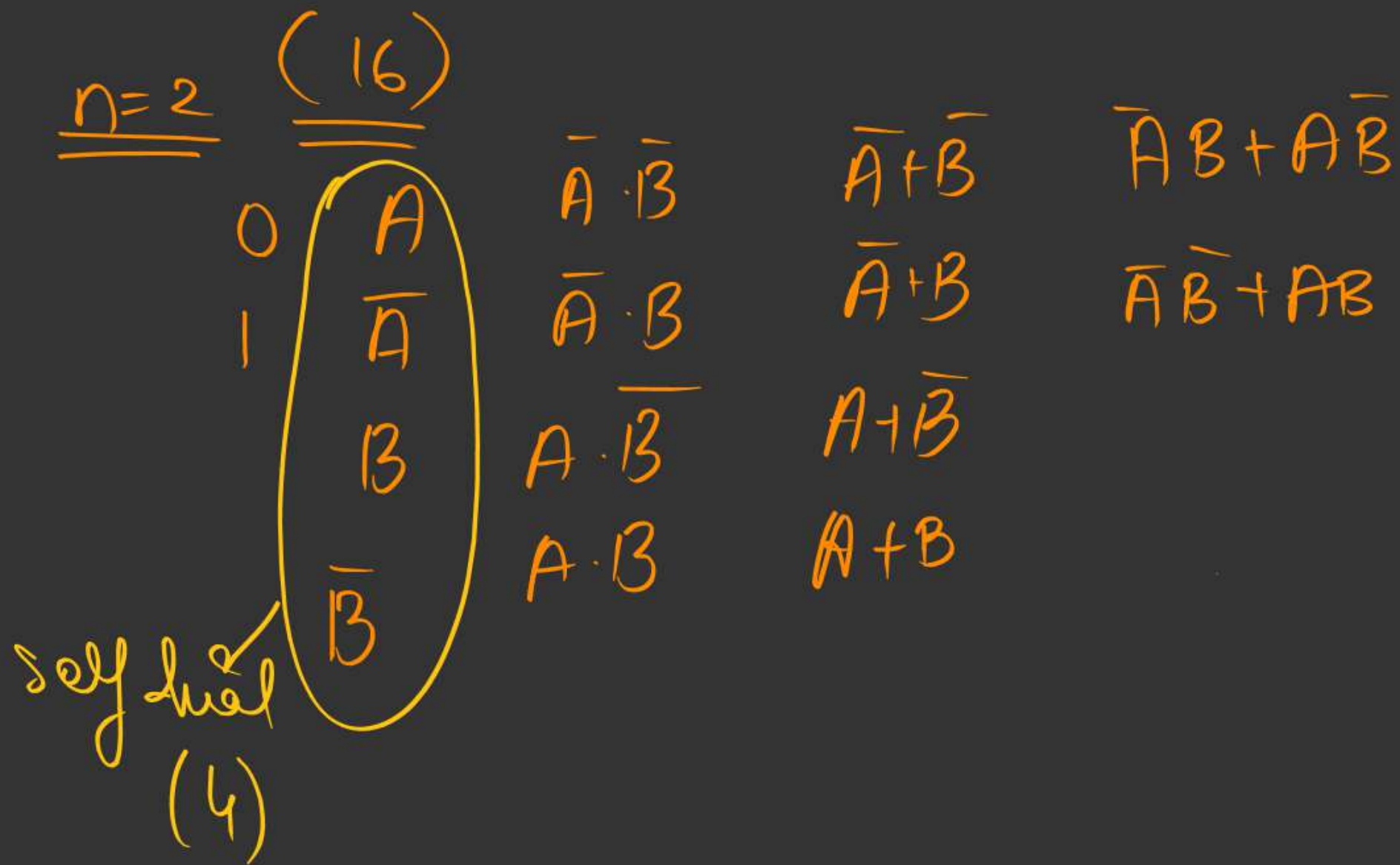
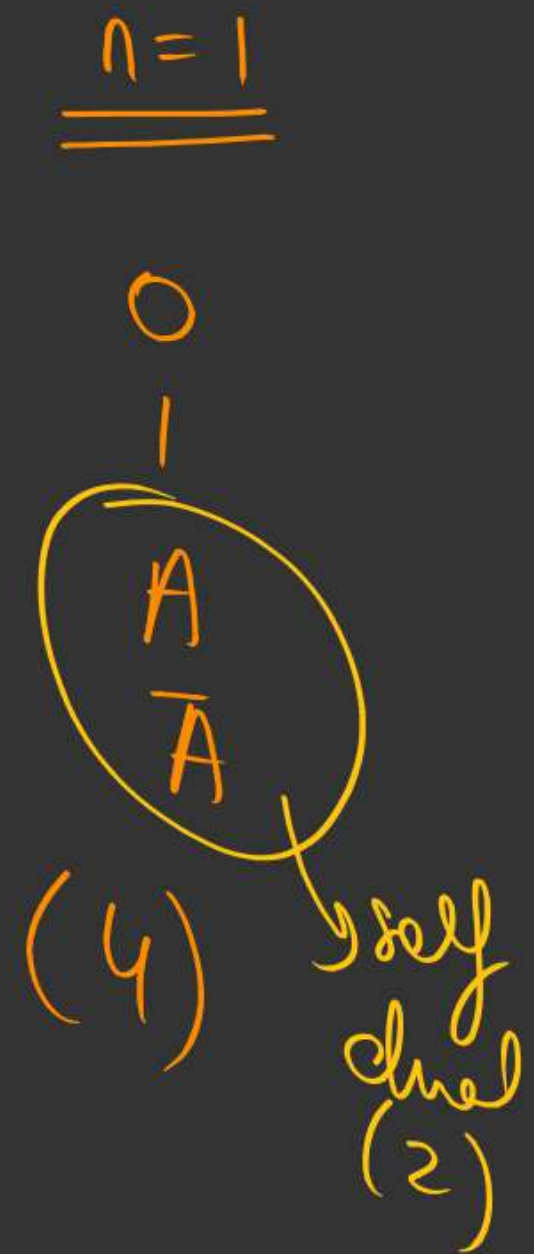
$$\rightarrow F = PQ + RS$$

$$F^D = (P+Q) \cdot (R+S)$$



$$\rightarrow (F^D)^D = P \cdot Q + R \cdot S$$

unique expressions $\rightarrow$ Simplified



Generalised

n variables



$$(2^{2^n})$$

expressions



$$2^{2^{n-1}}$$

Self-Dual expressions.

* Combinational Circuits :

↳ A circuit without
any feedback or memory

↳ Static circuit

5 steps Framework to Design a Combinational Ckt:

Step 1: Find the number of i/p's and o/p's. 

Step 2: Write the Truth Table.

Step 3: Write the logical expression.

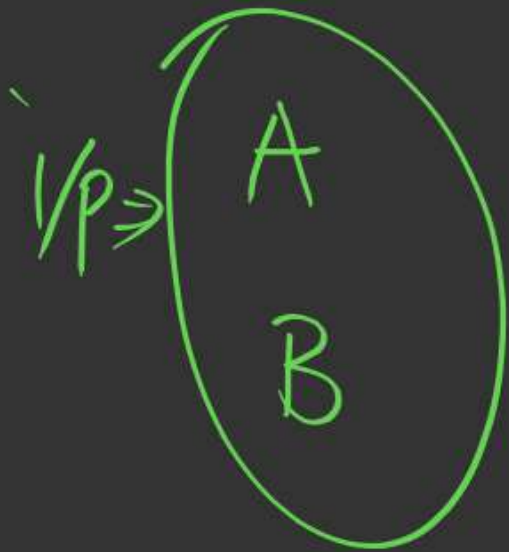
Step 4: Minimize the logical expression.

Step 5: Design the ckt / Implement the Hardware.

eg 1

Magnitude Comparator

1) Design a one-bit Comparator.



O/p $\Rightarrow$ 1) $A > B$

2) $A < B$

3) $A = B$



Step 2 :- Truth Table

(True $\rightarrow$ 1)
False $\rightarrow$ 0)

A	B	$(A > B)$ P	$(A < B)$ Q	$(A = B)$ R
0	0	0	0	1
0	1	0	<u>1</u>	0
1	0	<u>1</u>	0	0
1	1	0	0	1

Step 3 :- Logical Expression.

$$1) P(A > B) = A \bar{B}$$

$$2) Q(A < B) = \bar{A} B$$

$$3) R(A = B) = \bar{A} \bar{B} + AB = A \odot B$$



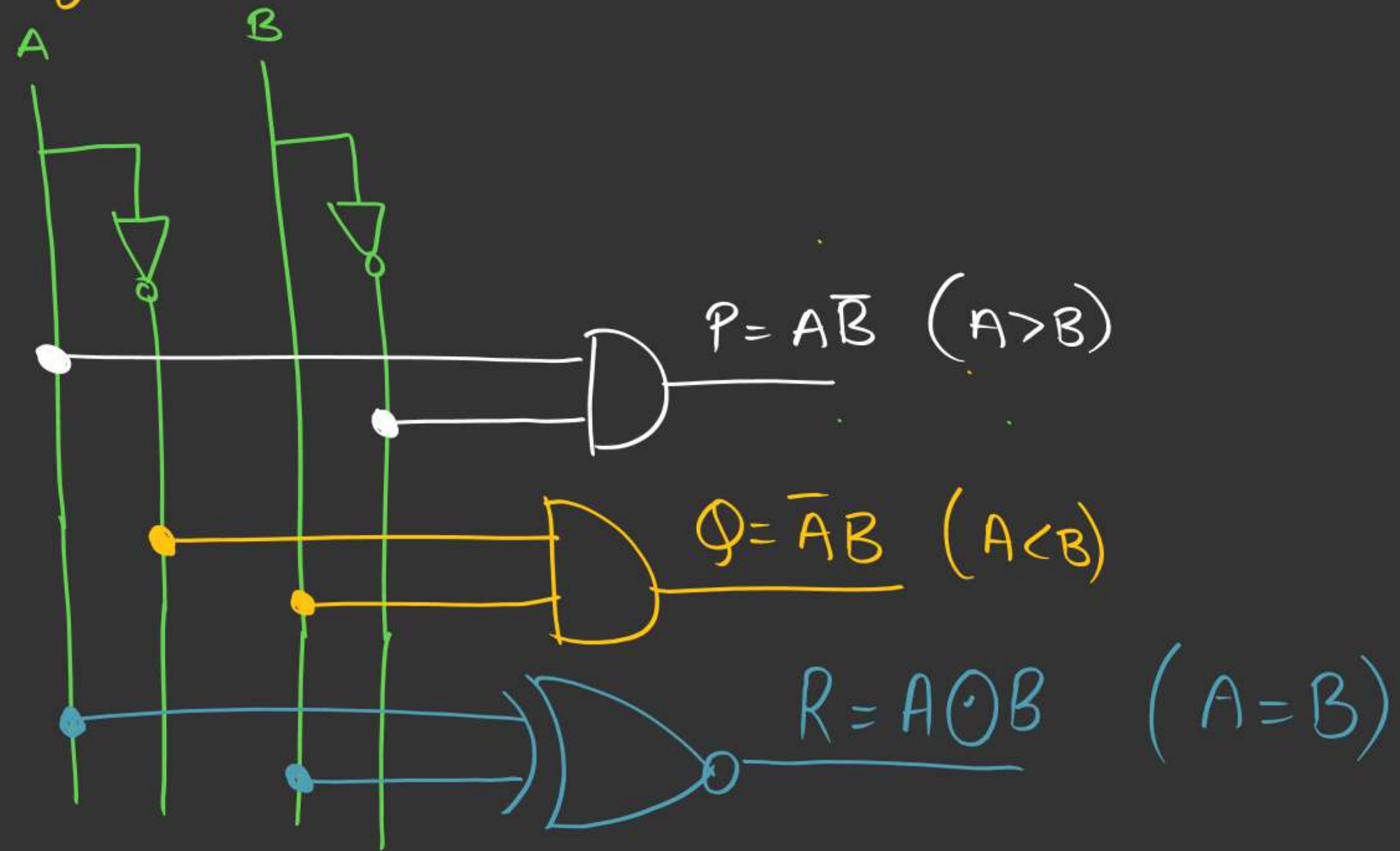
Step 4: minimize the expressions
↳ already minimized

Steps:- Logical diagram/Hardware Impl.

$$P = A\bar{B}$$

$$Q = \bar{A}B$$

$$R = A \oplus B$$



P.W

$$f(P, Q, R)$$

$f \Rightarrow \text{High}$, when atleast 1 of the
i/p is High.

→ Draw the K-map for f .

Let, No. of PI = 'a'
and No. of EPI = 'b'

then $a+b = ?$



2 mins Summary



- Completed K-map
- Duality
- Intro to Combination circuits

Tomorrow: 1 pm

3:30

} Backlog: HW, K-map with variables



THANK - YOU

